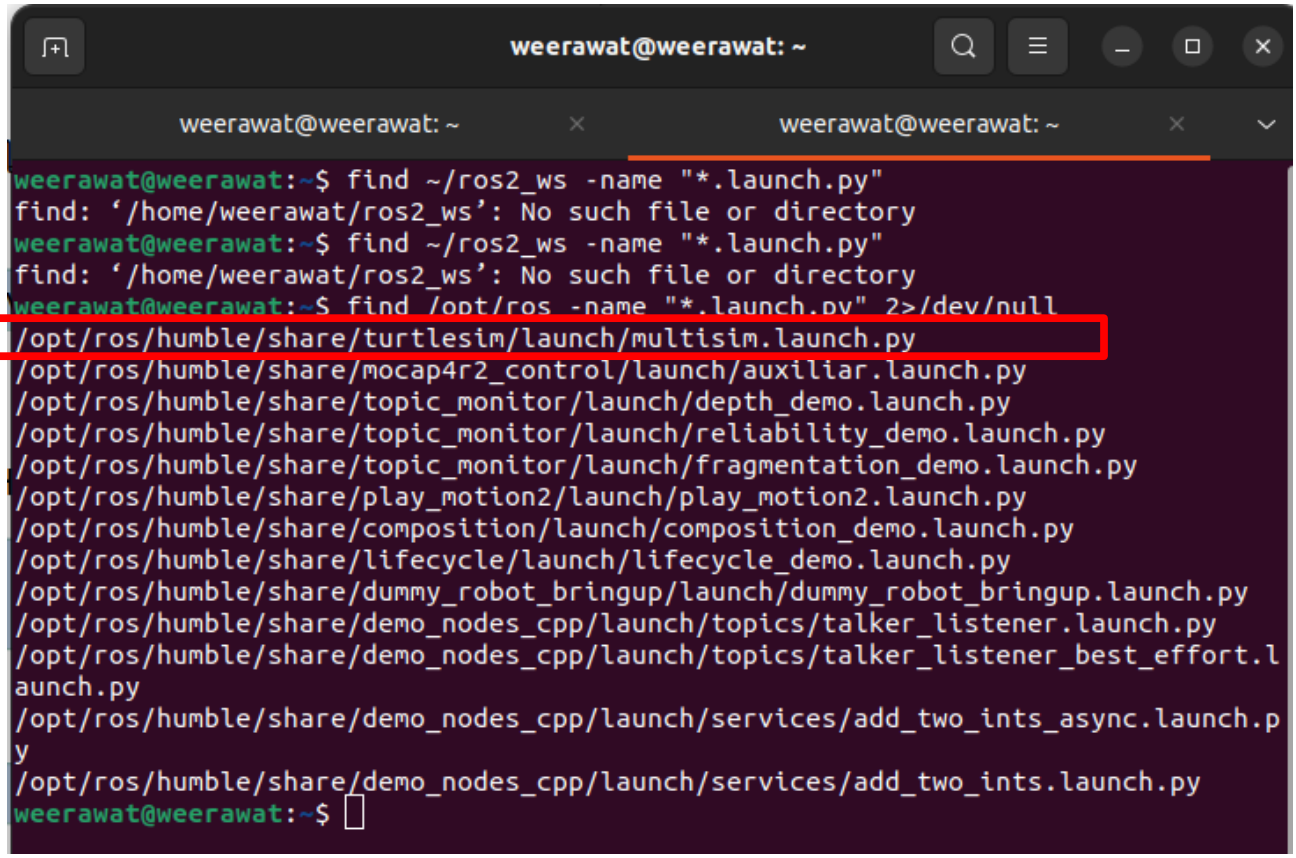
Launch File with publishing topics

1.Setup

ตรวจสอบตำแหน่งของ launch file เพื่อจะได้เข้าไปแก้ไขไฟล์ได้ (จะแสดง .launch.py ทั้งหมดภายใน ros)

```
find /opt/ros -name "*.launch.py" 2>/dev/null
```



จะแสดงตำแหน่งของ launch file multisim.launch.py

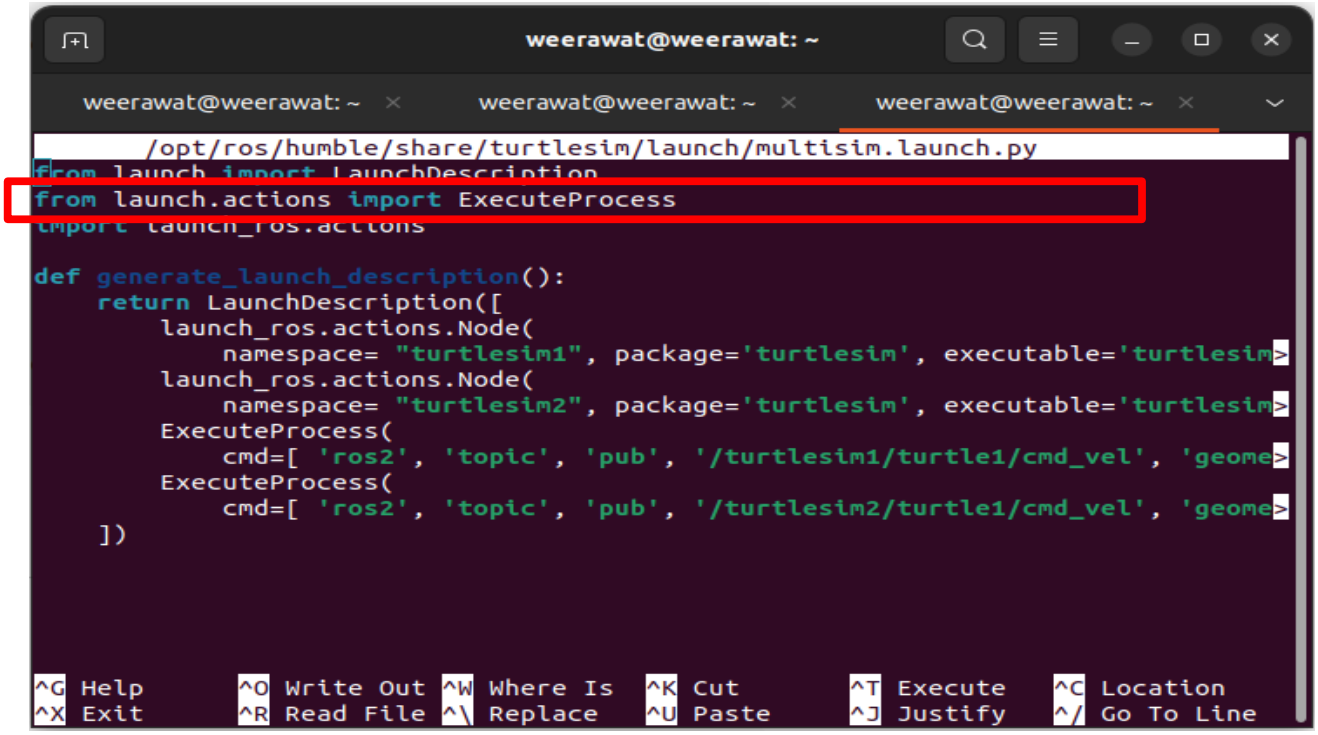
2.Edit launch file

จำเป็นต้องใช้ sudo เนื่องจากโฟเดอร์เป็น system directory การแก้ไขปกติจึงไม่สามารถใช้งานได้

```
sudo nano
/opt/ros/humble/share/turtlesim/launch/multisim.launch.py
```

import action สำหรับคำสั่งใน launch.py

```
from launch.actions import ExecuteProcess
```



เนื่องจาก ExecuteProcess คือ action ใน ros2 launch ที่ใช้สั่ง run คำสั่งโดยตรงของ topic

เพิ่มคำสั่งเพื่อให้ turtlesim หมุนพร้อมกันเมื่อ run launch file

```
        ExecuteProcess(
            cmd=[ 'ros2', 'topic', 'pub', '/turtlesim1/turtle1/cmd_vel',
'geometry_msgs/msg/Twist', '{linear: {x: 2.0}, angular: {z: 1.8}}', '-r',
'10'], output='screen'),
        ExecuteProcess(
            cmd=[ 'ros2', 'topic', 'pub', '/turtlesim2/turtle1/cmd_vel',
'geometry_msgs/msg/Twist', '{linear: {x: 2.0}, angular: {z: -1.8}}', '-r',
'10'], output='screen'),
```

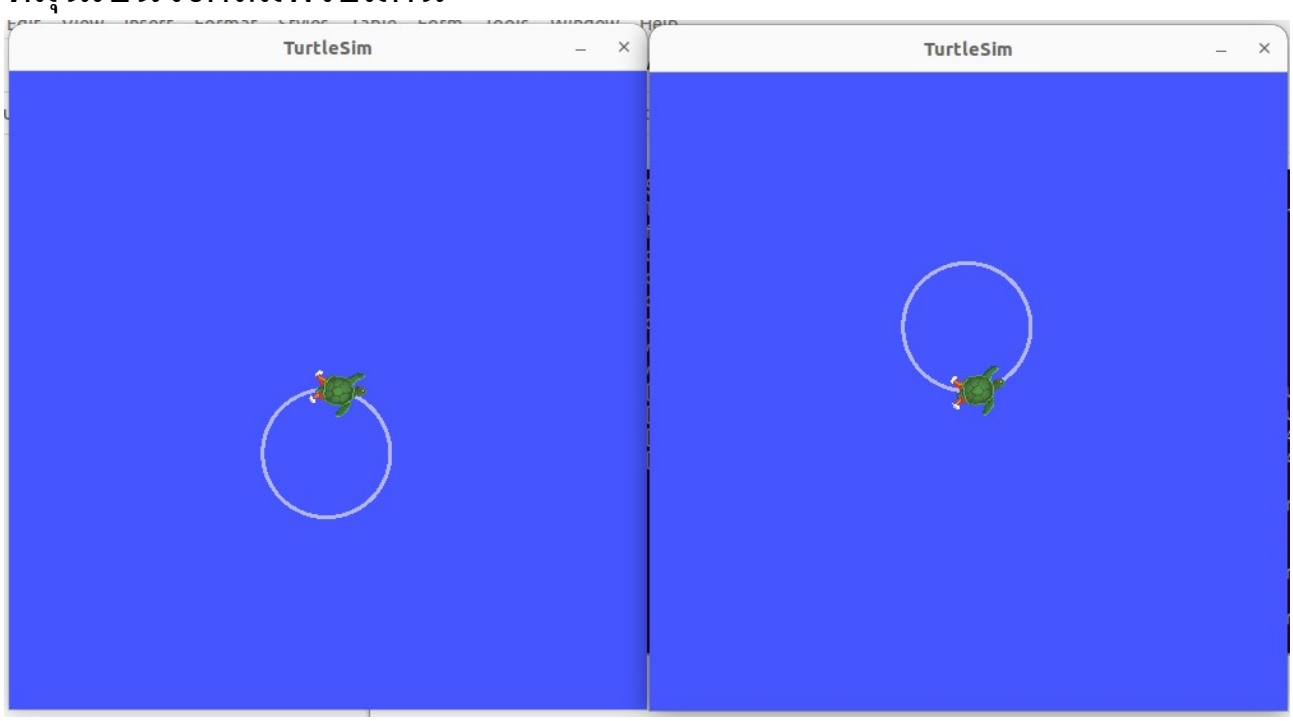
```
GNU nano 6.2                                    /opt/ros/humble/share/turtlesim/launch/multisim.launch.py
from launch import LaunchDescription
from launch.actions import ExecuteProcess
import launch_ros.actions

def generate_launch_description():
    return LaunchDescription([
        launch_ros.actions.Node(
            namespace= "turtlesim1", package='turtlesim', executable='turtlesim_node', output='screen'),
        launch_ros.actions.Node(
            namespace= "turtlesim2", package='turtlesim', executable='turtlesim_node', output='screen'),
        ExecuteProcess(
            cmd=[ 'ros2', 'topic', 'pub', '/turtlesim1/turtle1/cmd_vel', 'geometry_msgs/msg/Twist', '{linear: {x: 2.0}, angular: {z: 1.8}}', '-r', '10'], output='screen'),
        ExecuteProcess(
            cmd=[ 'ros2', 'topic', 'pub', '/turtlesim2/turtle1/cmd_vel', 'geometry_msgs/msg/Twist', '{linear: {x: 2.0}, angular: {z: -1.8}}', '-r', '10'], output='screen'),
    ])
```

3.Start Launch file

เริ่มต้นการทำงานของ launch file

```
ros2 launch turtlesim multisim.launch.py
```

จะปรากฏหน้าต่างการจำลอง และมีเต่าอยู่ตรงกลาง 2 หน้าต่าง ซึ่งจะหมุนเป็นวงกลมพร้อมกัน




```
weerawat@weerawat:~$ sudo nano /opt/ros/humble/share/turtlesim/launch/multisim.launch.py
[sudo] password for weerawat:
weerawat@weerawat:~$ ros2 launch turtlesim multisim.launch.py
[INFO] [launch]: All log files can be found below /home/weerawat/.ros/log/2026-01-04-18-50-58-089129-weerawat-9983
[INFO] [launch]: Default logging verbosity is set to INFO
[INFO] [turtlesim_node-1]: process started with pid [9984]
[INFO] [turtlesim_node-2]: process started with pid [9986]
[INFO] [ros2-3]: process started with pid [9988]
[INFO] [ros2-4]: process started with pid [9990]
[turtlesim_node-1] Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
[turtlesim_node-2] Warning: Ignoring XDG_SESSION_TYPE=wayland on Gnome. Use QT_QPA_PLATFORM=wayland to run on Wayland anyway.
[turtlesim_node-2] [INFO] [1767527458.435370206] [turtlesim2.turtlesim]: Starting turtlesim with node name /turtlesim2/turtlesim
[turtlesim_node-1] [INFO] [1767527458.435592958] [turtlesim1.turtlesim]: Starting turtlesim with node name /turtlesim1/turtlesim
[turtlesim_node-2] [INFO] [1767527458.441869166] [turtlesim2.turtlesim]: Spawning turtle [turtle1] at x=[5.544445], y=[5.544445], theta=[0.000000]
[turtlesim_node-1] [INFO] [1767527458.442402740] [turtlesim1.turtlesim]: Spawning turtle [turtle1] at x=[5.544445], y=[5.544445], theta=[0.000000]
[ros2-4] publisher: beginning loop
[ros2-4] publishing #1: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=2.0, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=1.8))
[ros2-4]
[ros2-3] publisher: beginning loop
[ros2-3] publishing #1: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=2.0, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=1.8))
[ros2-3]
[ros2-4] publishing #2: geometry_msgs.msg.Twist(linear=geometry_msgs.msg.Vector3(x=2.0, y=0.0, z=0.0), angular=geometry_msgs.msg.Vector3(x=0.0, y=0.0, z=1.8))
[ros2-4]
```